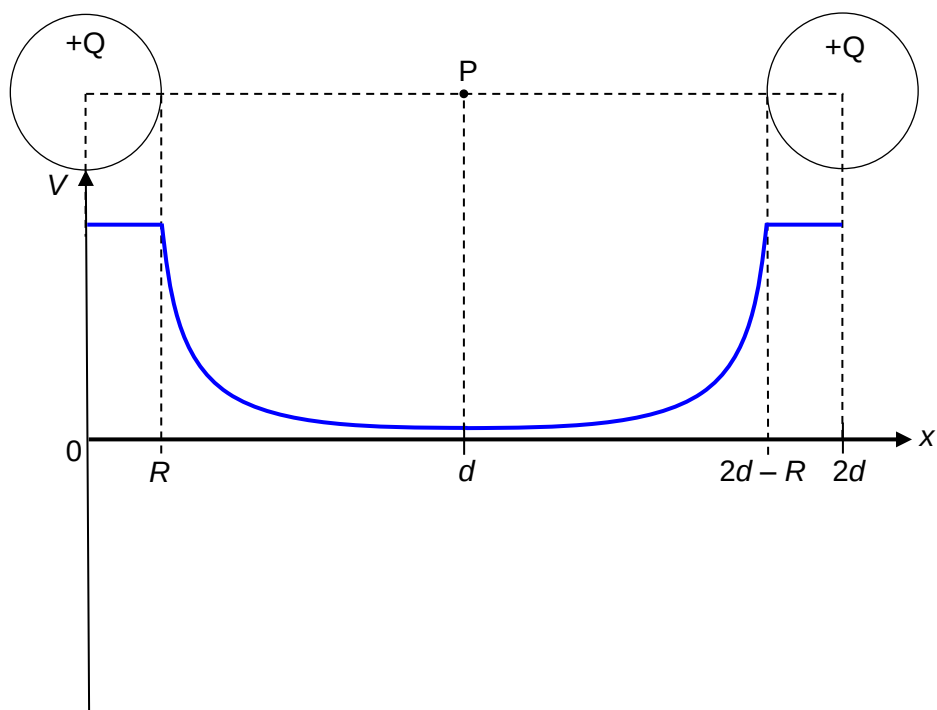
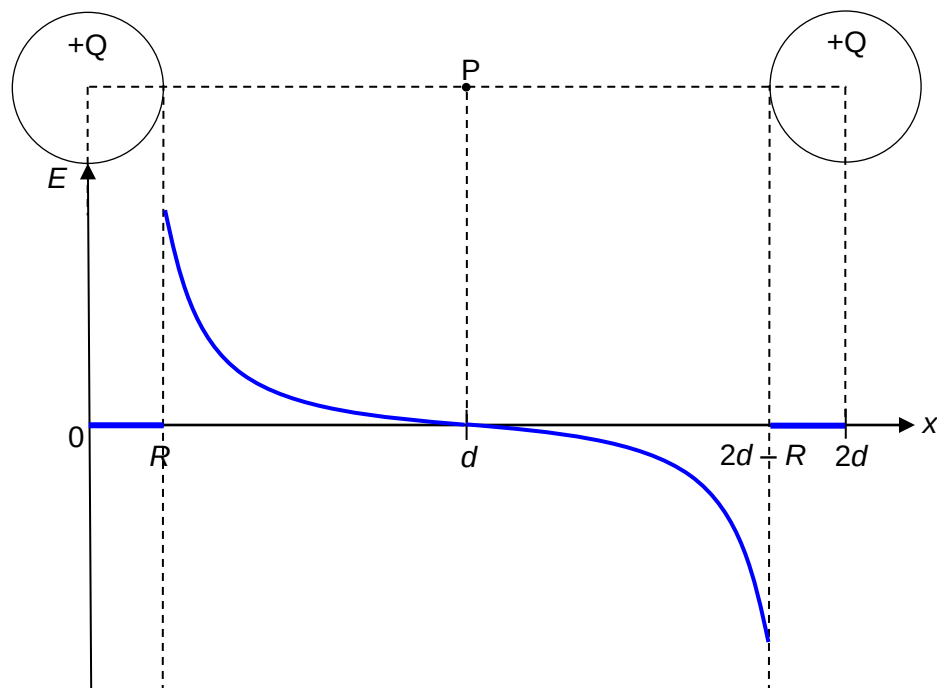


3 (a) (i)



(ii)



(b) (i) Solution 1

At point P, the electron is equidistant from each sphere and since each sphere carry the same amount of positive charge, the electric force acting on the electron due to sphere A is equal in magnitude and opposite in direction to the electric force on the electron due to sphere B.

Hence, the resultant electric force acting on the electron is zero.

(the force due to sphere A, $F_A = \frac{Qe}{4\pi\epsilon_0 d^2}$ is acting to the left and the force due to

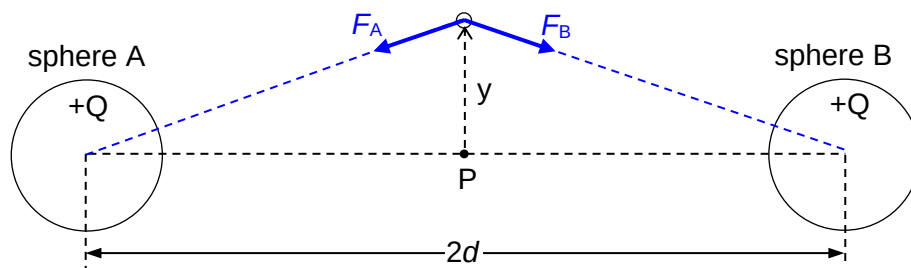
sphere B, $F_B = \frac{Qe}{4\pi\epsilon_0 d^2}$ is acting to the right)

Solution 2

Point P is equidistant from each sphere and since each sphere carry the same amount of positive charge, the electric field strength at point P due to sphere A is equal in magnitude and opposite in direction to the electric field strength due to sphere B. Hence, the resultant electric field strength at point P is zero.

Since the resultant electric force on the electron at point P is the product of the resultant electric field strength at point P and the elementary charge, the resultant electric force is zero.

(ii) 1.



2. The resultant force on the electron is acting vertically downwards, opposite to its upward displacement y .

$$F_R = -F_A \cos \theta - F_B \cos \theta$$

where θ is the angle that F_A or F_B makes with the vertical

$$\text{Since } F_A = F_B = \frac{Qe}{4\pi\epsilon_0} \frac{1}{(\sqrt{d^2 + y^2})^2} = \frac{Qe}{4\pi\epsilon_0(d^2 + y^2)}$$

$$\begin{aligned} F_R &= -2F_A \cos \theta \\ &= -2 \frac{Qe}{4\pi\epsilon_0(d^2 + y^2)} \left(\frac{y}{\sqrt{d^2 + y^2}} \right) \\ &= -\frac{Qe}{2\pi\epsilon_0} \left(\frac{y}{(d^2 + y^2)^{\frac{3}{2}}} \right) \end{aligned}$$

3. Since Q, e, m_e, d and ϵ_0 are constants, $a \propto (-y)$.

The acceleration a of the electron is directly proportional to its displacement y from the equilibrium position P.

The negative sign implies that the acceleration a acts in the opposite direction to the displacement y from the equilibrium position P. Hence, the acceleration points towards the equilibrium position.

This satisfies the condition for simple harmonic motion where $a = -\omega^2 y$. The electron will oscillate in simple harmonic motion along the vertical axis through point P with angular frequency $\omega = \sqrt{\frac{Qe}{2\pi\epsilon_0 m_e d^3}}$.

Raffles Institution
Year 5-6 Physics Department

4 (a) (i) Using Fleming's left hand rule, the magnetic force acting on the electron is always